

Sri Lankan Black Pepper

District-wise Regional Cultivation Guide

Knowledge Domain: 01 — Planting Practices (District Suitability)

Project: RAG Chat System for Sri Lankan Black Pepper Knowledge

This document provides comprehensive district-level knowledge for black pepper (*Piper nigrum*) cultivation across all 25 districts of Sri Lanka. For each district, it documents the agro-ecological zone, elevation range, annual rainfall, soil types, and the recommended pepper varieties suited to those specific conditions.

1. Province and District Overview

Sri Lanka is divided into 9 provinces covering 25 districts. Black pepper cultivation potential varies significantly across these regions based on rainfall, elevation, and soil composition.

Province Summary:

Province	Districts Covered	Primary Agro Zone
Central	Kandy, Matale, Nuwara Eliya	Wet Zone
Sabaragamuwa	Ratnapura, Kegalle	Wet Zone
Uva	Badulla, Monaragala	Intermediate
Western	Colombo, Gampaha, Kalutara	Wet Zone
Southern	Galle, Matara, Hambantota	Wet Zone
North Western	Kurunegala, Puttalam	Intermediate
North Central	Anuradhapura, Polonnaruwa	Dry Zone
Northern	Jaffna, Kilinochchi, Mullaitivu, Vavuniya, Mannar	Dry Zone
Eastern	Trincomalee, Batticaloa, Ampara	Dry Zone

2. Master District Reference Table

The table below consolidates all 25 districts with their agro-ecological classification, physical characteristics, soil profiles, and recommended black pepper varieties for rapid reference.

All 25 Districts — Cultivation Suitability Reference:

District	Province	Agro Zone	Elevation	Rainfall	Soil Types	Recommended Varieties
Kandy	Central	Wet Zone / Mid-Country	500–1,500 m	2,000–3,000 mm/yr	Loamy, Clay Loam, Red-Yellow Podzolic	GK 49, Kohukumbure Rala, Dingi Rala
Matale	Central	Intermediate / Mid-Country	300–900 m	1,500–2,500 mm/yr	Loamy, Clay Loam, Red-Yellow Podzolic	Panniyur-1, GK 49, Dingi Rala
Nuwara Eliya	Central	Wet Zone / Up-Country	1,200–2,500 m	2,500–5,000 mm/yr	Loamy, Clay Loam, Red-Yellow Podzolic	Kohukumbure Rala, Dingi Rala, GK 49
Ratnapura	Sabaragamuwa	Wet Zone / Low-Country	30–500 m	3,000–5,000 mm/yr	Loamy, Clay Loam, Red-Yellow Podzolic	Panniyur-1, GK 49, Kuching
Kegalle	Sabaragamuwa	Wet Zone / Low-Mid Country	50–700 m	2,500–4,000 mm/yr	Loamy, Clay Loam, Red-Yellow Podzolic	Panniyur-1, GK 49
Badulla	Uva	Intermediate / Up-Country	600–2,000 m	1,500–2,500 mm/yr	Loamy, Clay Loam, Red-Yellow Podzolic, Gravelly	Bootawe Rala, GK 49, Dingi Rala
Monaragala	Uva	Dry Zone / Low-Country	60–500 m	1,200–1,800 mm/yr	Gravelly, Loamy	Kuching, GK 49
Colombo	Western	Wet Zone / Low-Country	0–50 m	2,400–2,800 mm/yr	Loamy, Sandy	Panniyur-1, Kuching
Gampaha	Western	Wet Zone / Low-Country	0–100 m	2,000–2,800 mm/yr	Loamy, Sandy	Panniyur-1, Kuching, GK 49
Kalutara	Western	Wet Zone / Low-Country	0–100 m	2,200–3,500 mm/yr	Loamy, Sandy	Panniyur-1, Kuching, GK 49
Galle	Southern	Wet Zone / Low-Country	0–200 m	2,000–3,000 mm/yr	Loamy, Sandy	Panniyur-1, Kuching, GK 49
Matara	Southern	Wet Zone / Low-Country	0–200 m	1,800–2,500 mm/yr	Loamy, Sandy	Panniyur-1, Kuching, GK 49
Hambantota	Southern	Dry Zone / Low-Country	0–150 m	900–1,500 mm/yr	Sandy, Gravelly	Kuching, GK 49
Kurunegala	North Western	Intermediate / Low-Country	50–300 m	1,200–2,000 mm/yr	Gravelly, Loamy	Kuching, GK 49

District	Province	Agro Zone	Elevation	Rainfall	Soil Types	Recommended Varieties
Puttalam	North Western	Dry Zone / Low-Country	0–100 m	800–1,200 mm/yr	Sandy, Gravelly	Kuching, GK 49
Anuradhapura	North Central	Dry Zone / Low-Country	50–200 m	1,000–1,600 mm/yr	Gravelly, Loamy	Kuching, GK 49
Polonnaruwa	North Central	Dry Zone / Low-Country	50–200 m	1,100–1,700 mm/yr	Gravelly, Loamy	Kuching, GK 49
Jaffna	Northern	Dry Zone / Low-Country	0–100 m	900–1,200 mm/yr	Sandy, Gravelly	Kuching, GK 49
Kilinochchi	Northern	Dry Zone / Low-Country	0–80 m	900–1,300 mm/yr	Sandy, Gravelly	Kuching, GK 49
Mullaitivu	Northern	Dry Zone / Low-Country	0–80 m	900–1,400 mm/yr	Sandy, Gravelly	Kuching, GK 49
Vavuniya	Northern	Dry Zone / Low-Country	50–150 m	1,000–1,600 mm/yr	Sandy, Gravelly	Kuching, GK 49
Mannar	Northern	Dry Zone / Low-Country	0–50 m	700–1,100 mm/yr	Sandy, Gravelly	Kuching, GK 49
Trincomalee	Eastern	Dry Zone / Low-Country	0–200 m	1,200–1,800 mm/yr	Sandy, Gravelly	Kuching, GK 49
Batticaloa	Eastern	Dry Zone / Low-Country	0–100 m	1,500–2,000 mm/yr	Sandy, Gravelly	Kuching, GK 49
Ampara	Eastern	Intermediate / Low-Country	0–300 m	1,500–2,200 mm/yr	Sandy, Gravelly	Kuching, GK 49

3. Soil Type Reference and Implications

Five primary soil types are found across Sri Lanka's black pepper cultivation zones. Understanding each soil's characteristics helps determine appropriate variety selection, fertilizer management, and irrigation strategy.

Soil Type	Description & Implications for Black Pepper
Loamy	Balanced mix of sand, silt, and clay. Best water retention and drainage for pepper. Found across all zones.
Clay Loam	Higher clay content. Good nutrient holding capacity. Common in mid and up-country elevations. Requires drainage management.
Red-Yellow Podzolic	Highly weathered tropical soils. Naturally acidic. Widespread in wet and intermediate zones. Requires lime/dolomite to correct pH.
Gravelly	Coarse-textured with free drainage. Common in dry-intermediate zones. Low moisture retention — mulching is essential.
Sandy	High permeability, low nutrient and water retention. Found in coastal and dry zone districts. Heavy organic matter inputs required.

Key Insight: Red-Yellow Podzolic soils are the most widespread in Sri Lanka's established black pepper regions (Central, Sabaragamuwa, Uva provinces). These soils require pH correction with Dolomite (500 g/plant every 2 years) as described in the Maintenance knowledge domain.

4. DEA Recommended Hybrid Varieties

The Department of Export Agriculture (DEA) of Sri Lanka has recently introduced three new hybrid black pepper varieties — Dingi Rala, Bootawe Rala, and Kohukumbure Rala. These hybrids were developed from crosses of high-performing parents and are now available for commercial cultivation. Note: IW 5 is no longer an active variety in Sri Lanka and has been replaced by these DEA hybrids in district recommendations throughout this document.

Important Update: IW 5 has been discontinued as a recommended variety in Sri Lanka. Farmers previously growing or planning to plant IW 5 should switch to the appropriate DEA hybrid based on their district. Additionally, GK 49 is recognised as Sri Lanka's superior local selection (not a separate 'Local' variety) — all district recommendations now correctly reflect GK 49 in this role.

DEA Hybrid Varieties — Technical Specifications:

Specification	Dingi Rala	Bootawe Rala	Kohukumbure Rala
Parent Varieties	Panniyur-1 × GK 49	Panniyur-1 × DM 7	MW 21 × Panniyur-1

Specification	Dingi Rala	Bootawe Rala	Kohukumbure Rala
Panicle Length	12 cm	14 cm	12 cm
Filling Percentage	80%	80%	80%
Yield (g/vine/year)	2,245 g	2,724 g ★ Highest	2,340 g
Dry Weight Ratio	327:1	327:1	246:1
Oleoresin %	12.9%	12.9%	15.4% ★ Highest
Oil %	2.8%	3.1%	3.6% ★ Highest
Piperine %	5.6%	6.3% ★ Highest	6.0%
Weight of 1 Litre	571.86 g	490.88 g	613.43 g
Weight of 1,000 Seeds	63.44 g	47.19 g	53.31 g
Best Use Case	High yield; wet/intermediate zones	Max yield; up-country intermediate	Export quality; high oil & oleoresin

Selection Guide: Choose *Bootawe Rala* for maximum yield. Choose *Kohukumbure Rala* for premium export quality (highest oil and oleoresin content). Choose *Dingi Rala* for a balanced, well-performing hybrid in wet and intermediate zones.

5. Variety Suitability by Agro-Ecological Zone

The matrix below maps all current recommended varieties — including the three DEA hybrids — against Sri Lanka's three primary agro-ecological zones to guide region-specific planting decisions. (★ denotes a DEA hybrid variety)

Variety	Wet Zone	Intermediate Zone	Dry Zone	Notes
Panniyur-1	✓ Highly Suitable	✓ Suitable	✗ Not Recommended	World's first pepper hybrid; vigorous grower, requires high humidity; best in low to mid-country wet zones
Kuching	✓ Suitable	✓ Suitable	✓ Suitable	Most adaptable variety; high piperine content; tolerates drier and adverse conditions across all zones
GK 49	✓ Highly Suitable	✓ Suitable	~ Marginal	High-yielding local selection with very high pungency; best in red-yellow podzolic soils of wet/intermediate zones
GK 49 (Local Sel.)	✓ Highly Suitable	✓ Suitable	~ Marginal	Sri Lanka's premier high-yielding local selection; superior pungency

Variety	Wet Zone	Intermediate Zone	Dry Zone	Notes
				and quality; widely grown across wet and intermediate zones
Dingi Rala ★	✓ Highly Suitable	✓ Highly Suitable	✗ Not Recommended	DEA hybrid (Panniyur-1 × GK 49); 2,245g/vine/yr; 12cm panicle; 5.6% piperine; suited to wet and intermediate zones
Bootawe Rala ★	✓ Highly Suitable	✓ Highly Suitable	✗ Not Recommended	DEA hybrid (Panniyur-1 × DM 7); highest yield at 2,724g/vine/yr; 14cm panicle; 6.3% piperine; top pick for up-country intermediate zones
Kohukumbure Rala ★	✓ Highly Suitable	✓ Suitable	✗ Not Recommended	DEA hybrid (MW 21 × Panniyur-1); 2,340g/vine/yr; highest oleoresin (15.4%) and oil (3.6%); suited for mid to up-country wet zones

Note: *Kuching is the most adaptable variety across all three zones and is the default primary for dry zone districts. GK 49 — Sri Lanka's premier local selection — is recommended across wet and intermediate zones. DEA hybrids (Dingi Rala, Bootawe Rala, Kohukumbure Rala) are recommended for wet and intermediate zone districts seeking maximum yield and quality.*

6. District Variety Priority Ranking

The table below provides a definitive ranked recommendation for every district. Each district is assigned a Primary variety (best match for local conditions and highest commercial yield), a Secondary variety (suitable but requires more inputs or yields slightly less), and an Alternative variety (tolerates conditions but is best suited for subsistence or small-scale farming). Where only two varieties are suitable, no Alternative is listed.

■ Primary — Highest yield, best soil/climate fit, recommended for commercial planting. ■ Secondary — Suitable but needs more inputs or has a slight yield trade-off. ■ Alternative — Tolerates conditions; best for subsistence or mixed farming.

District	 Primary	 Secondary	 Alternative	Zone	Selection Reasoning
Kandy	GK 49	Kohukumbure Rala	Dingi Rala	Wet Zone	GK 49 is the leading local selection for Kandy's mid-country Red-Yellow Podzolic soils, offering superior pungency. Kohukumbure Rala (highest oleoresin

District	 Primary	 Secondary	 Alternative	Zone	Selection Reasoning
					15.4%) is the recommended DEA hybrid. Dingi Rala is a well-balanced alternative hybrid.
Matale	Panniyur-1	GK 49	Dingi Rala	Intermediate	Panniyur-1 thrives in Matale's open intermediate zone with high humidity. GK 49, as Sri Lanka's top local selection, offers high pungency and yield as a strong secondary. Dingi Rala (Panniyur-1 × GK 49 hybrid) is a productive alternative.
Nuwara Eliya	Kohukumbure Rala	Dingi Rala	GK 49	Wet Zone	With IW 5 no longer available, Kohukumbure Rala is the best replacement for up-country elevations. Dingi Rala is a strong secondary hybrid. GK 49, Sri Lanka's premier local selection, serves as the alternative where hybrid planting material is unavailable.
Ratnapura	Panniyur-1	GK 49	Kuching	Wet Zone	Ratnapura's high rainfall and humid wet-zone conditions are ideal for Panniyur-1's vigorous growth. GK 49 excels in its red-yellow podzolic soils. Kuching handles wetter spells well.
Kegalle	Panniyur-1	GK 49	—	Wet Zone	Panniyur-1 is the top performer in Kegalle's wet low-mid-country zone. GK 49, Sri Lanka's superior local selection, provides high pungency and yield as an excellent secondary choice.
Badulla	Bootawe Rala	GK 49	Dingi Rala	Intermediate	Bootawe Rala (highest yield at

District	 Primary	 Secondary	 Alternative	Zone	Selection Reasoning
					2,724g/vine/year) is ideal for Badulla's intermediate up-country zone. GK 49, the superior local selection, performs well in gravelly loamy pockets. Dingi Rala is a productive hybrid alternative.
Monaragala	Kuching	GK 49	—	Dry Zone	Kuching's drought tolerance makes it the top choice for Monaragala's dry conditions. GK 49, as the premier local selection, is viable in loamy pockets with supplemental irrigation and serves as a productive secondary.
Colombo	Panniyur-1	Kuching	—	Wet Zone	Panniyur-1 flourishes in Colombo's wet coastal climate. Kuching is suitable where sandy soils dominate. Limited space in urban Colombo constrains variety choices.
Gampaha	Panniyur-1	Kuching	GK 49	Wet Zone	Panniyur-1 is highly productive in Gampaha's wet zone loamy soils. Kuching suits sandier plots. GK 49, Sri Lanka's high-yielding local selection, is a strong alternative in loamy areas.
Kalutara	Panniyur-1	Kuching	GK 49	Wet Zone	High rainfall in Kalutara suits Panniyur-1 best. Kuching is reliable on sandy coastal soils. GK 49, Sri Lanka's top local selection, is an excellent alternative on loamy inland plots.

District	 Primary	 Secondary	 Alternative	Zone	Selection Reasoning
Galle	Panniyur-1	Kuching	GK 49	Wet Zone	Galle's wet-zone humidity and loamy soils strongly favour Panniyur-1. Kuching handles sandy coastal patches well. GK 49 is an excellent alternative on loamy inland soils for farmers seeking a high-pungency local selection.
Matara	Panniyur-1	Kuching	GK 49	Wet Zone	Similar conditions to Galle; Panniyur-1 is the top performer. Kuching handles the sandy patches. GK 49 is the recommended local selection as an alternative in loamy areas.
Hambantota	Kuching	GK 49	—	Dry Zone	Kuching is the primary choice for Hambantota's harsh dry conditions. GK 49, being a drought-tolerant local selection, is the recommended secondary where supplemental irrigation is available.
Kurunegala	Kuching	GK 49	—	Intermediate	Kuching's adaptability to gravelly soils and drier spells makes it the primary choice. GK 49, Sri Lanka's premier local selection, is productive in loamy pockets and is the clear secondary choice.
Puttalam	Kuching	GK 49	—	Dry Zone	Puttalam's low rainfall and sandy soils make Kuching the primary commercial variety. GK 49 is viable with supplemental irrigation on loamy patches as the

District	 Primary	 Secondary	 Alternative	Zone	Selection Reasoning
					secondary local selection.
Anuradhapura	Kuching	GK 49	—	Dry Zone	Kuching is the most drought-tolerant commercial variety for Anuradhapura. GK 49, as the superior local selection, performs well in loamy areas with supplemental irrigation.
Polonnaruwa	Kuching	GK 49	—	Dry Zone	Kuching's resilience in dry gravelly soils makes it the primary commercial choice. GK 49 is the recommended local selection as a secondary option where loamy soils or irrigation are available.
Jaffna	Kuching	GK 49	—	Dry Zone	Jaffna's low rainfall and sandy soils make Kuching the primary choice. GK 49 can be grown as a secondary with drip irrigation. Supplemental irrigation is strongly advised for any commercial planting.
Kilinochchi	Kuching	GK 49	—	Dry Zone	Conditions are similar to Jaffna. Kuching is the primary commercial option. GK 49 is recommended as the secondary local selection with supplemental irrigation.
Mullaitivu	Kuching	GK 49	—	Dry Zone	Sandy coastal soils and low rainfall limit variety choices. Kuching is the most tolerant to adverse conditions. GK 49 serves as the secondary local

District	 Primary	 Secondary	 Alternative	Zone	Selection Reasoning
					selection with irrigation support.
Vavuniya	Kuching	GK 49	—	Dry Zone	Vavuniya has slightly better rainfall than coastal northern districts. Kuching remains the primary choice. GK 49 is the recommended local selection as secondary where irrigation is available.
Mannar	Kuching	GK 49	—	Dry Zone	Mannar has the lowest annual rainfall in Sri Lanka. Kuching is the only primary commercial choice. GK 49 may be grown as a secondary with drip irrigation and heavy mulching.
Trincomalee	Kuching	GK 49	—	Dry Zone	Trincomalee receives bimodal rainfall from both monsoons. Kuching is the primary commercial variety. GK 49 is the recommended local selection as secondary where soil conditions allow.
Batticaloa	Kuching	GK 49	—	Dry Zone	Batticaloa's coastal sandy soils favour Kuching as primary. Its higher rainfall compared to other eastern districts makes GK 49 a viable secondary local selection.
Ampara	Kuching	GK 49	—	Intermediate	Ampara's intermediate zone and slightly higher rainfall allow GK 49 as a secondary option. Kuching remains primary due to soil compatibility. Local is a low-risk alternative.

How to use this table: For a farmer in any Sri Lankan district, identify the row for their district and plant the Primary variety for maximum commercial output. If planting material for the Primary is unavailable, choose Secondary. Use the Alternative only when the other two are inaccessible or for mixed home gardens.

7. District-by-District Profiles

The following section provides a structured profile for each of the 25 districts, grouped by province. Each profile lists the specific soil types, agro-ecological zone, elevation, rainfall, and recommended black pepper varieties for that district.

1. Central Province (3 Districts)

■ Kandy — Wet Zone / Mid-Country

Province	Central
Agro-Ecological Zone	Wet Zone / Mid-Country
Elevation Range	500–1,500 m
Annual Rainfall	2,000–3,000 mm/yr
Soil Types	Loamy, Clay Loam, Red-Yellow Podzolic
Primary Variety	GK 49
Secondary Variety	Kohukumbure Rala
Alternative Variety	Dingi Rala
Selection Reasoning	GK 49 is the leading local selection for Kandy's mid-country Red-Yellow Podzolic soils, offering superior pungency. Kohukumbure Rala (highest oleoresin 15.4%) is the recommended DEA hybrid. Dingi Rala is a well-balanced alternative hybrid.

■ Matale — Intermediate / Mid-Country

Province	Central
Agro-Ecological Zone	Intermediate / Mid-Country
Elevation Range	300–900 m
Annual Rainfall	1,500–2,500 mm/yr
Soil Types	Loamy, Clay Loam, Red-Yellow Podzolic
Primary Variety	Panniyur-1
Secondary Variety	GK 49
Alternative Variety	Dingi Rala

Selection Reasoning	Panniyur-1 thrives in Matale's open intermediate zone with high humidity. GK 49, as Sri Lanka's top local selection, offers high pungency and yield as a strong secondary. Dingi Rala (Panniyur-1 × GK 49 hybrid) is a productive alternative.
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■ Nuwara Eliya — Wet Zone / Up-Country

Province	Central
Agro-Ecological Zone	Wet Zone / Up-Country
Elevation Range	1,200–2,500 m
Annual Rainfall	2,500–5,000 mm/yr
Soil Types	Loamy, Clay Loam, Red-Yellow Podzolic
Primary Variety	Kohukumbure Rala
Secondary Variety	Dingi Rala
Alternative Variety	GK 49
Selection Reasoning	With IW 5 no longer available, Kohukumbure Rala is the best replacement for up-country elevations. Dingi Rala is a strong secondary hybrid. GK 49, Sri Lanka's premier local selection, serves as the alternative where hybrid planting material is unavailable.

2. Sabaragamuwa Province (2 Districts)

■ Ratnapura — Wet Zone / Low-Country

Province	Sabaragamuwa
Agro-Ecological Zone	Wet Zone / Low-Country
Elevation Range	30–500 m
Annual Rainfall	3,000–5,000 mm/yr
Soil Types	Loamy, Clay Loam, Red-Yellow Podzolic
Primary Variety	Panniyur-1
Secondary Variety	GK 49
Alternative Variety	Kuching
Selection Reasoning	Ratnapura's high rainfall and humid wet-zone conditions are ideal for Panniyur-1's vigorous growth. GK 49 excels in its red-yellow podzolic soils. Kuching handles wetter spells well.

■ **Kegalle** — *Wet Zone / Low-Mid Country*

Province	Sabaragamuwa
Agro-Ecological Zone	Wet Zone / Low-Mid Country
Elevation Range	50–700 m
Annual Rainfall	2,500–4,000 mm/yr
Soil Types	Loamy, Clay Loam, Red-Yellow Podzolic
Primary Variety	Panniyur-1
Secondary Variety	GK 49
Alternative Variety	—
Selection Reasoning	Panniyur-1 is the top performer in Kegalle's wet low-mid-country zone. GK 49, Sri Lanka's superior local selection, provides high pungency and yield as an excellent secondary choice.

3. Uva Province (2 Districts)

■ **Badulla** — *Intermediate / Up-Country*

Province	Uva
Agro-Ecological Zone	Intermediate / Up-Country
Elevation Range	600–2,000 m
Annual Rainfall	1,500–2,500 mm/yr
Soil Types	Loamy, Clay Loam, Red-Yellow Podzolic, Gravelly
Primary Variety	Bootawe Rala
Secondary Variety	GK 49
Alternative Variety	Dingi Rala
Selection Reasoning	Bootawe Rala (highest yield at 2,724g/vine/year) is ideal for Badulla's intermediate up-country zone. GK 49, the superior local selection, performs well in gravelly loamy pockets. Dingi Rala is a productive hybrid alternative.

■ **Monaragala** — *Dry Zone / Low-Country*

Province	Uva
Agro-Ecological Zone	Dry Zone / Low-Country
Elevation Range	60–500 m
Annual Rainfall	1,200–1,800 mm/yr

Soil Types	Gravelly, Loamy
Primary Variety	Kuching
Secondary Variety	GK 49
Alternative Variety	—
Selection Reasoning	Kuching's drought tolerance makes it the top choice for Monaragala's dry conditions. GK 49, as the premier local selection, is viable in loamy pockets with supplemental irrigation and serves as a productive secondary.

4. Western Province (3 Districts)

■ Colombo — Wet Zone / Low-Country

Province	Western
Agro-Ecological Zone	Wet Zone / Low-Country
Elevation Range	0–50 m
Annual Rainfall	2,400–2,800 mm/yr
Soil Types	Loamy, Sandy
Primary Variety	Panniyur-1
Secondary Variety	Kuching
Alternative Variety	—
Selection Reasoning	Panniyur-1 flourishes in Colombo's wet coastal climate. Kuching is suitable where sandy soils dominate. Limited space in urban Colombo constrains variety choices.

■ Gampaha — Wet Zone / Low-Country

Province	Western
Agro-Ecological Zone	Wet Zone / Low-Country
Elevation Range	0–100 m
Annual Rainfall	2,000–2,800 mm/yr
Soil Types	Loamy, Sandy
Primary Variety	Panniyur-1
Secondary Variety	Kuching
Alternative Variety	GK 49

Selection Reasoning	Panniyur-1 is highly productive in Gampaha's wet zone loamy soils. Kuching suits sandier plots. GK 49, Sri Lanka's high-yielding local selection, is a strong alternative in loamy areas.
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■ **Kalutara** — *Wet Zone / Low-Country*

Province	Western
Agro-Ecological Zone	Wet Zone / Low-Country
Elevation Range	0–100 m
Annual Rainfall	2,200–3,500 mm/yr
Soil Types	Loamy, Sandy
Primary Variety	Panniyur-1
Secondary Variety	Kuching
Alternative Variety	GK 49
Selection Reasoning	High rainfall in Kalutara suits Panniyur-1 best. Kuching is reliable on sandy coastal soils. GK 49, Sri Lanka's top local selection, is an excellent alternative on loamy inland plots.

5. Southern Province (3 Districts)

■ **Galle** — *Wet Zone / Low-Country*

Province	Southern
Agro-Ecological Zone	Wet Zone / Low-Country
Elevation Range	0–200 m
Annual Rainfall	2,000–3,000 mm/yr
Soil Types	Loamy, Sandy
Primary Variety	Panniyur-1
Secondary Variety	Kuching
Alternative Variety	GK 49
Selection Reasoning	Galle's wet-zone humidity and loamy soils strongly favour Panniyur-1. Kuching handles sandy coastal patches well. GK 49 is an excellent alternative on loamy inland soils for farmers seeking a high-pungency local selection.

■ **Matara** — *Wet Zone / Low-Country*

Province	Southern
Agro-Ecological Zone	Wet Zone / Low-Country
Elevation Range	0–200 m
Annual Rainfall	1,800–2,500 mm/yr
Soil Types	Loamy, Sandy
Primary Variety	Panniyur-1
Secondary Variety	Kuching
Alternative Variety	GK 49
Selection Reasoning	Similar conditions to Galle; Panniyur-1 is the top performer. Kuching handles the sandy patches. GK 49 is the recommended local selection as an alternative in loamy areas.

■ **Hambantota** — *Dry Zone / Low-Country*

Province	Southern
Agro-Ecological Zone	Dry Zone / Low-Country
Elevation Range	0–150 m
Annual Rainfall	900–1,500 mm/yr
Soil Types	Sandy, Gravelly
Primary Variety	Kuching
Secondary Variety	GK 49
Alternative Variety	—
Selection Reasoning	Kuching is the primary choice for Hambantota's harsh dry conditions. GK 49, being a drought-tolerant local selection, is the recommended secondary where supplemental irrigation is available.

6. North Western Province (2 Districts)

■ **Kurunegala** — *Intermediate / Low-Country*

Province	North Western
Agro-Ecological Zone	Intermediate / Low-Country
Elevation Range	50–300 m
Annual Rainfall	1,200–2,000 mm/yr
Soil Types	Gravelly, Loamy

Primary Variety	Kuching
Secondary Variety	GK 49
Alternative Variety	—
Selection Reasoning	Kuching's adaptability to gravelly soils and drier spells makes it the primary choice. GK 49, Sri Lanka's premier local selection, is productive in loamy pockets and is the clear secondary choice.

■ **Puttalam** — *Dry Zone / Low-Country*

Province	North Western
Agro-Ecological Zone	Dry Zone / Low-Country
Elevation Range	0–100 m
Annual Rainfall	800–1,200 mm/yr
Soil Types	Sandy, Gravelly
Primary Variety	Kuching
Secondary Variety	GK 49
Alternative Variety	—
Selection Reasoning	Puttalam's low rainfall and sandy soils make Kuching the primary commercial variety. GK 49 is viable with supplemental irrigation on loamy patches as the secondary local selection.

7. North Central Province (2 Districts)

■ **Anuradhapura** — *Dry Zone / Low-Country*

Province	North Central
Agro-Ecological Zone	Dry Zone / Low-Country
Elevation Range	50–200 m
Annual Rainfall	1,000–1,600 mm/yr
Soil Types	Gravelly, Loamy
Primary Variety	Kuching
Secondary Variety	GK 49
Alternative Variety	—
Selection Reasoning	Kuching is the most drought-tolerant commercial variety for Anuradhapura. GK 49, as the superior local selection, performs well in loamy areas with supplemental irrigation.

■ Polonnaruwa — Dry Zone / Low-Country

Province	North Central
Agro-Ecological Zone	Dry Zone / Low-Country
Elevation Range	50–200 m
Annual Rainfall	1,100–1,700 mm/yr
Soil Types	Gravelly, Loamy
Primary Variety	Kuching
Secondary Variety	GK 49
Alternative Variety	—
Selection Reasoning	Kuching's resilience in dry gravelly soils makes it the primary commercial choice. GK 49 is the recommended local selection as a secondary option where loamy soils or irrigation are available.

8. Northern Province (5 Districts)

■ Jaffna — Dry Zone / Low-Country

Province	Northern
Agro-Ecological Zone	Dry Zone / Low-Country
Elevation Range	0–100 m
Annual Rainfall	900–1,200 mm/yr
Soil Types	Sandy, Gravelly
Primary Variety	Kuching
Secondary Variety	GK 49
Alternative Variety	—
Selection Reasoning	Jaffna's low rainfall and sandy soils make Kuching the primary choice. GK 49 can be grown as a secondary with drip irrigation. Supplemental irrigation is strongly advised for any commercial planting.

■ Kilinochchi — Dry Zone / Low-Country

Province	Northern
Agro-Ecological Zone	Dry Zone / Low-Country
Elevation Range	0–80 m

Annual Rainfall	900–1,300 mm/yr
Soil Types	Sandy, Gravelly
Primary Variety	Kuching
Secondary Variety	GK 49
Alternative Variety	—
Selection Reasoning	Conditions are similar to Jaffna. Kuching is the primary commercial option. GK 49 is recommended as the secondary local selection with supplemental irrigation.

■ Mullaitivu — Dry Zone / Low-Country

Province	Northern
Agro-Ecological Zone	Dry Zone / Low-Country
Elevation Range	0–80 m
Annual Rainfall	900–1,400 mm/yr
Soil Types	Sandy, Gravelly
Primary Variety	Kuching
Secondary Variety	GK 49
Alternative Variety	—
Selection Reasoning	Sandy coastal soils and low rainfall limit variety choices. Kuching is the most tolerant to adverse conditions. GK 49 serves as the secondary local selection with irrigation support.

■ Vavuniya — Dry Zone / Low-Country

Province	Northern
Agro-Ecological Zone	Dry Zone / Low-Country
Elevation Range	50–150 m
Annual Rainfall	1,000–1,600 mm/yr
Soil Types	Sandy, Gravelly
Primary Variety	Kuching
Secondary Variety	GK 49
Alternative Variety	—
Selection Reasoning	Vavuniya has slightly better rainfall than coastal northern districts. Kuching remains the primary choice. GK 49 is the recommended local selection as secondary where irrigation is available.

■ Mannar — Dry Zone / Low-Country

Province	Northern
Agro-Ecological Zone	Dry Zone / Low-Country
Elevation Range	0–50 m
Annual Rainfall	700–1,100 mm/yr
Soil Types	Sandy, Gravelly
Primary Variety	Kuching
Secondary Variety	GK 49
Alternative Variety	—
Selection Reasoning	Mannar has the lowest annual rainfall in Sri Lanka. Kuching is the only primary commercial choice. GK 49 may be grown as a secondary with drip irrigation and heavy mulching.

9. Eastern Province (3 Districts)

■ Trincomalee — Dry Zone / Low-Country

Province	Eastern
Agro-Ecological Zone	Dry Zone / Low-Country
Elevation Range	0–200 m
Annual Rainfall	1,200–1,800 mm/yr
Soil Types	Sandy, Gravelly
Primary Variety	Kuching
Secondary Variety	GK 49
Alternative Variety	—
Selection Reasoning	Trincomalee receives bimodal rainfall from both monsoons. Kuching is the primary commercial variety. GK 49 is the recommended local selection as secondary where soil conditions allow.

■ Batticaloa — Dry Zone / Low-Country

Province	Eastern
Agro-Ecological Zone	Dry Zone / Low-Country
Elevation Range	0–100 m
Annual Rainfall	1,500–2,000 mm/yr

Soil Types	Sandy, Gravelly
Primary Variety	Kuching
Secondary Variety	GK 49
Alternative Variety	—
Selection Reasoning	Batticaloa's coastal sandy soils favour Kuching as primary. Its higher rainfall compared to other eastern districts makes GK 49 a viable secondary local selection.

■ **Ampara** — *Intermediate / Low-Country*

Province	Eastern
Agro-Ecological Zone	Intermediate / Low-Country
Elevation Range	0–300 m
Annual Rainfall	1,500–2,200 mm/yr
Soil Types	Sandy, Gravelly
Primary Variety	Kuching
Secondary Variety	GK 49
Alternative Variety	—
Selection Reasoning	Ampara's intermediate zone and slightly higher rainfall allow GK 49 as a secondary option. Kuching remains primary due to soil compatibility. Local is a low-risk alternative.